

Loops in Vb.net

Instructor Name: Assist.Lec. Aseel F. Alshaibanee

Instructor Name: Assist.Lec. Marwa Ali

Département: Computer Science

Level: Two

Loop Structure

- Loops are used to repeat statements until a condition is met.
- Types:
 - For...Next
 - While
 - Do Until
 - Do While

For...Next Loop

- Used when the number of iterations is known.
- Includes:
 - Start value
 - End value
 - Step
- For...Next Example

```
For i = 1 To 10 Step 2
    MessageBox.Show(i)
Next
```
- Output: 1,3,5,7,9

Practical Example (Odd/Even)

- Enter 10 numbers
- If $\text{number} \bmod 2 = 0 \rightarrow \text{Even}$
- Else $\rightarrow \text{Odd}$
- Display results in ComboBox

Practical Example (Odd/Even)

The image shows a screenshot of a Windows application window titled "Form1". The window has a standard Windows title bar with minimize, maximize, and close buttons. The main content area is divided into two columns. The left column is headed "The Odd Numbers" and contains a text box with a dropdown arrow. The right column is headed "The Even Numbers" and also contains a text box with a dropdown arrow. At the bottom of the window, there are three buttons: "Find", "Clear", and "Exit". The "Exit" button is currently selected, as indicated by a dashed border and small square handles around it.

Practical Example (Odd/Even)

```
Public Class Form1
    0 references
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Dim X As Integer
        Dim I As Integer
        For I = 1 To 10
            X = InputBox("Enter The 10 Numbers")
            If X Mod 2 = 0 Then
                ComboBox2.Items.Add(X)
            Else
                ComboBox1.Items.Add(X)
            End If
        Next
    End Sub

    0 references
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        ComboBox1.Items.Clear()
        ComboBox2.Items.Clear()
    End Sub

    0 references
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        End
    End Sub
End Class
```


While Loop

- Used when number of iterations is unknown.
- Runs while condition is True
- While Example

`i = 1`

`While i <= 10`

`MessageBox.Show(i)`

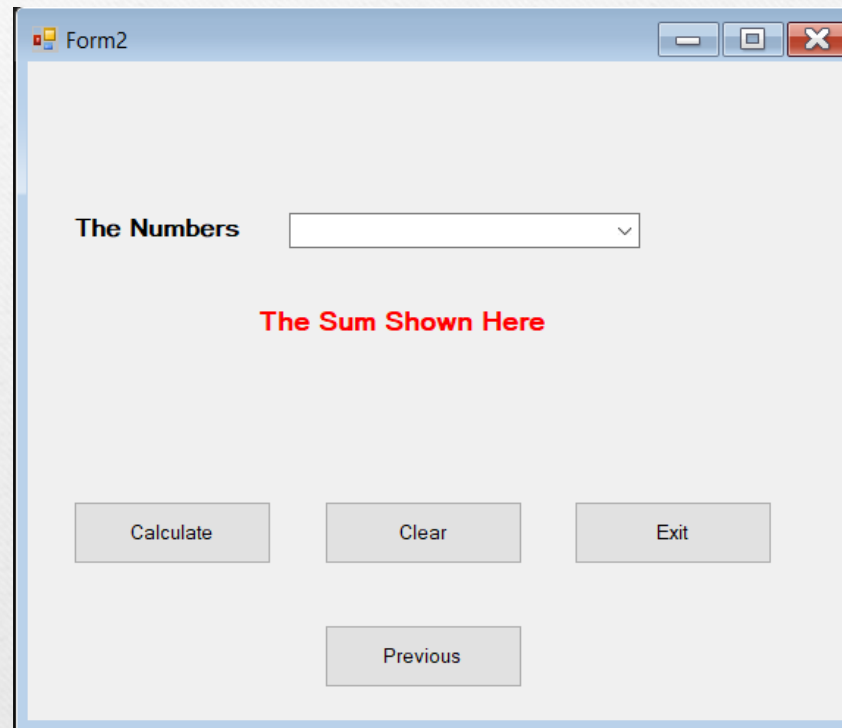
`i = i + 1`

`End While`

Practical While Example

- Enter 5 Numbers
- Display all Numbers in Combobox
- Sum the Even Numbers
- Display the result of Summation in Label

Practical While Example



The screenshot shows a Windows application window titled "Form2". Inside the window, there is a label "The Numbers" followed by a dropdown menu. Below this, the text "The Sum Shown Here" is displayed in red. At the bottom of the window, there are four buttons: "Calculate", "Clear", "Exit", and "Previous".

The Numbers

The Sum Shown Here

Calculate Clear Exit

Previous

Practical While Example

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    Dim X As Integer
    Dim I As Integer
    Dim Sum As Integer = 0
    I = 1
    While I <= 5
        X = InputBox("Enter The 5 Numbers")
        ComboBox1.Items.Add(X)
        If X Mod 2 = 0 Then
            Sum = Sum + X
        End If
        I = I + 1
    End While
    Label2.Text = Sum
End Sub
0 references
Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
    ComboBox1.Items.Clear()
    Label2.Text = ""
End Sub
0 references
Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
    End
End Sub
0 references
Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
    Form1.Show()
    Me.Hide()
End Sub
```


Do-While | Until

The syntax for the Do ... While | Until statement is as follows:

Do

Statement(s)

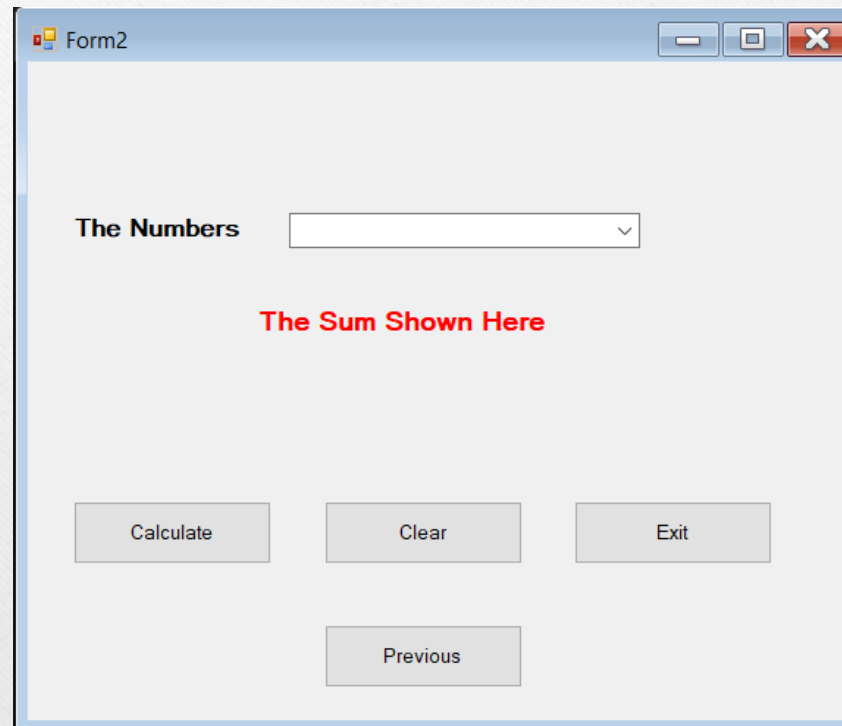
Exit Do

Loop While | Until Condition

Practical Example

- Enter 5 Numbers
- Display all Numbers in Combobox
- Sum the Even Numbers
- Display the result of Summation in Label

Practical Example



The image shows a screenshot of a Windows application window titled "Form2". The window has a standard Windows title bar with minimize, maximize, and close buttons. Inside the window, there is a label "The Numbers" followed by a dropdown menu. Below the dropdown menu, the text "The Sum Shown Here" is displayed in red. At the bottom of the window, there are four buttons: "Calculate", "Clear", "Exit", and "Previous".

The Numbers

The Sum Shown Here

Calculate Clear Exit

Previous

Practical Example

```
Public Class Form2
    0 references
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        Dim X As Integer
        Dim I As Integer
        Dim Sum As Integer = 0
        I = 1
        Do
            X = InputBox("Enter The 5 Numbers")
            ComboBox1.Items.Add(X)
            If X Mod 2 = 0 Then
                Sum = Sum + X
            End If
            I = I + 1
        Loop While I <= 5
        Label2.Text = Sum
    End Sub
    0 references
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        ComboBox1.Items.Clear()
        Label2.Text = ""
    End Sub
    0 references
    Private Sub Button3_Click(sender As Object, e As EventArgs) Handles Button3.Click
        End
    End Sub
    0 references
    Private Sub Button4_Click(sender As Object, e As EventArgs) Handles Button4.Click
        Form1.Show()
        Me.Hide()
    End Sub
End Class
```

Thank You